5.8.3 The Theory of Evolution and Natural Selection

After completing this chapter you should be able to relate the structure and function of living things to models, theories and laws.

For Questions 1 to 5 choose the best answer (A), (B), (C) or (D).

1. Consider this statement:

'Organisms who are able to survive well in an environment pass on their characteristics to more offspring and these characteristics become more common'.

What theory does this statement allude to?

(A) The theory of adaptations.
(B) The theory of genetic modification.
(C) The theory of natural selection.
(D) The theory of natural survival.

2. What causes variations in the genetic make-up of organisms from the same species?

(A) Mutations caused by external factors.
(B) Inherited characteristics from parents.
(C) Environmental factors.
(D) Both inherited and environmental factors.

3. What is the meaning of the term 'evolution'?

(A) To develop gradually.
(B) To adapt to one's surroundings.
(C) To change as mutations occur.
(D) To change as genes change.

4. Which of the following statements is *not* proposed by natural selection?

(A) Members within a single species show variation.
(B) Humans have slowly developed from apes.
(C) In the struggle to survive, some members of a species will die before they reproduce.
(D) Over time, organisms become better suited to their environment.

5. In what way does artificial selection provide evidence for natural selection?

(A) In deliberately isolating favoured characteristics in organisms and breeding new generations that contain those characteristics, it copies natural selection.
(B) It forces natural selection to occur over a much shorter period of time.
(C) It produces changes in the DNA of organisms by genetic engineering and this is similar to the way in which natural selection works.
(D) Artificial selection is a totally different process to natural selection and, as such, provides no evidence for natural selection.

6.

(a) Explain the concept of 'survival of the fittest'.

Members of a speciec will have variations in their characteristics. Some of these characteristics will help some individuals survive and reproduce in their environment bette than bthers. In this way, the individuals with the favourable characteristics will breed more and live longer and ove time the less favourable characterstics are bred out.

(b) What is another term for 'survival of the fittest'?

Natural selection.

7.

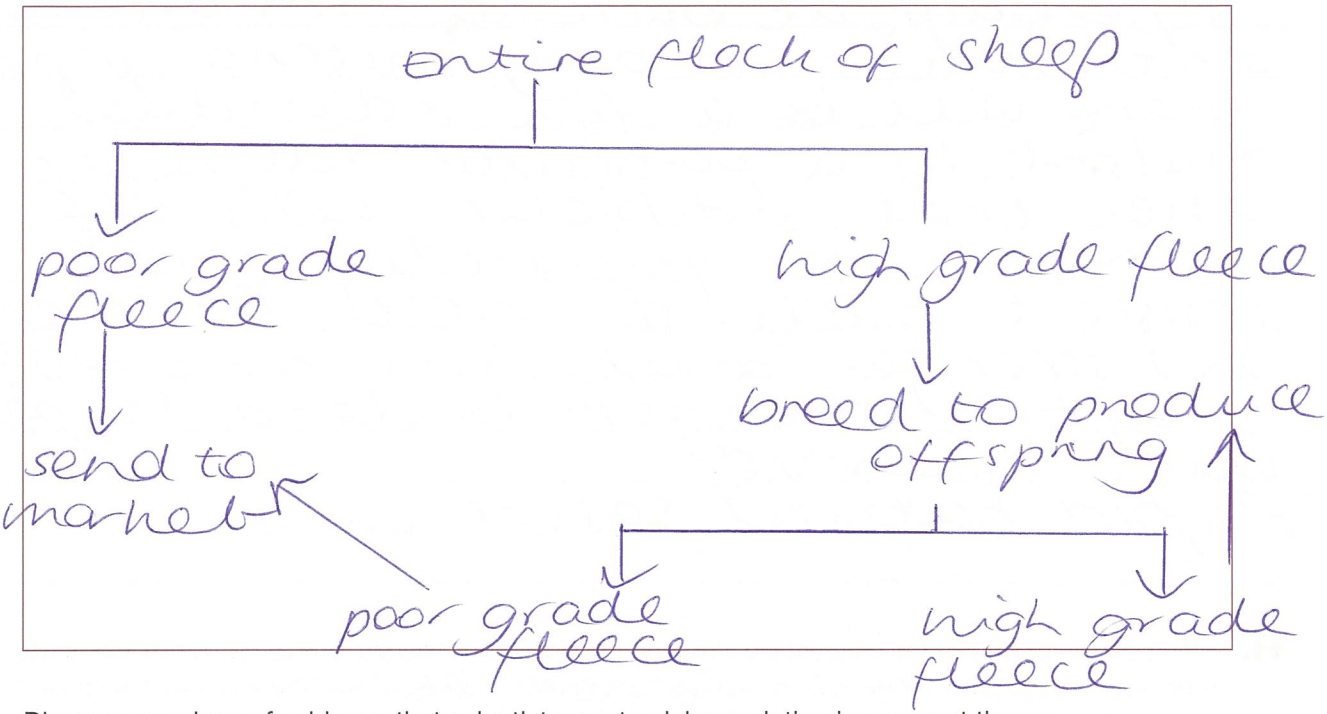
(a) How does *artificial selection* differ from *natural selection*?

In artificial selection humans select there individuals with favourable characterstics and deliberately breed them so that the favourable characterstic appears in the offspring. In natural selection selection pressures breed out unfavourable characterstics, only those with favourable characterstics survive.

(b) A farmer had a flock of sheep that gave mixed quality wool. Some gave very high grade fleece, others quite poor fleece. Explain how the farmer could use artificial selection to improve the overall quality of the fleece produced by his flock.

The farmer should select his breeding stock from the high grade fleece and keep breeding from the offspring that show high grade fleece.

(c) Convert your ideas from (b) into a flow chart to summarise the procedure you recommend.



8. Discuss one piece of evidence that scientists use to claim evolution is a correct theory.

Comparative anatomy
Comparative embryology
Biogeography
Fossil evidence
Molecular biology

9. Explain the importance of sexual reproduction in an evolutionary process.

Sexual reproduction creates variation due to the offspring being a combination of the parents genetic characteristics variation is the driving force for evolution. Asexual reproduction creates genetically identical offspring.

10. Explain how the development of super bugs supports Darwin's idea of 'the survival of the fittest'.

Superbugs are bacteria resistant to antibiotics - In any bacteria population there will be a few individuals resistant to a particular antibiotic. When that antibiotic kills off the rest of the population, the resistant bacteria breed. As stronger and stronger antibiotics are used, only the strongest bacteria 'super bugs' survive.
Rapid natural selection.

11. When Darwin examined the different life forms on the Galapagos Islands, one of his observations was that there was considerable variation in the size and shapes of the beaks of different species of finches.

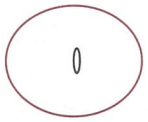
Explain how the theory of evolution would account for this variation.

Successful variations help these organisms to survive and so pass on these variations to their offspring. In the case of the finches the shapes of their beaks allow them to eat particular kinds of food and the survival of a particular beak shape will depend on the type of food available. Finches that have beaks adapted to feed well will survive better than those whose beaks cannot obtain food as easily in particular environments.

12. The diagrams show how natural selection works in a bacteria population.

A  Bacteria type X in culture medium.

B  Bacteria X multiply.
Some mutate to form new strain type Y.

C  Antibiotic added which kills all bacteria X.

D  Type Y bacteria multiply, unaffected by antibiotic.

(a) Explain how the information above is an example of natural selection.

Shows that only the antibiotic resistant bacteria survive to go on to produce a new population better adapted to survive the new conditions.

(b) When an organism reproduces, DNA mutations can occur. Explain how these can be helpful in the survival of the organism.

Mutations may result in characteristics that are favourable to the survival of the individual. These can be passed on to the offspring and their offspring. Over time, many members of the population will inherit the characteristics and so the species becomes better adapted to survive.

(c) Recall the name of the thread-like structures that contain DNA.

Chromosomes

(d) In which part of a cell are these structures found?

Nucleus

13. Describe the process of natural selection.

Process by which organisms with characteristics most suited for survival in a particular population in a particular environment increase in number as they pass on favoured characteristics to their offspring while those without the characteristic decrease and may become extinct

14. Outline the evidence for present-day organisms having developed from different organisms in the past.

- Fossils - show an orderly sequence of development from simple to more complex life forms.
- Comparative embryology - suggests
- Comparative anatomy - descent from a common ancestor
- Biogeography - indicates that on different continents organisms have developed independently